Inequalities — problem 7.19

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The Fibonacci sequence is defined by $a_1 = a_2 = 1$, $a_{n+2} = a_n + a_{n+1}$. Prove that

$$\frac{1}{2} + \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \frac{5}{2^5} + \cdots + \frac{a_n}{2^n} < 2.$$